# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 1: **Real Numbers** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

### SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The decimal expansion of $\frac{17}{8}$ terminates after how many places of decimals? 1

**(a)** 3 **(b)** 5 **(c)** 4 **(d)** 6

**2.** Which of the following is irrational? 1

**(a)** $\sqrt{4}+2$ **(b)** $\sqrt{7}+2$ **(c)** $\sqrt{49}-2$ **(d)** $2\times\sqrt{4}$

**3.** For which of the following is $n$ such that $7^n$ ends with the digit 0? 1

**(a)** no natural number $n$ **(b)** only even $n$

**(c)** $n=2$ only **(d)** every $n$

**4.** In Euclid's division lemma for 203 and 15, $203 = 15q + r$. The value of $q$ is: 1

**(a)** 12 **(b)** 14 **(c)** 13 **(d)** 8

**5.** The exponent of 7 in the prime factorisation of 1176 is: 1

**(a)** 3 **(b)** 4 **(c)** 5 **(d)** 2

**6.** If HCF(336, 54) = 6, then LCM(336, 54) is: 1

**(a)** 2016 **(b)** 18144 **(c)** 3024 **(d)** 390

**7.** The product of a non-zero rational and an irrational number such as $5\sqrt{2}$ is: 1

**(a)** sometimes rational **(b)** always rational

**(c)** always irrational **(d)** always an integer

**8.** The decimal expansion of $\frac{23}{200}$ terminates after how many places of decimals? 1

**(a)** 3 **(b)** 6 **(c)** 4 **(d)** 5

**9.** Which of the following is irrational? 1

**(a)** $\sqrt{9}-4$ **(b)** $\sqrt{16}+4$ **(c)** $4\times\sqrt{16}$ **(d)** $\sqrt{3}+4$

**10.** For which of the following is $n$ such that $9^n$ ends with the digit 0? 1

**(a)** only even $n$ **(b)** no natural number $n$

**(c)** $n=2$ only **(d)** every $n$

### SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Given that HCF(336,54)=6, find LCM(336,54). 2

**12.** Explain why $7\times11+7$ is a composite number. 2

**13.** Check whether $21^n$ can end with the digit 0 for any natural number $n$. 2

**14.** Given that HCF(510,92)=2, find LCM(510,92). 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Prove that $\sqrt{5}$ is irrational. 3

**16.** Show that $5 + 4\sqrt{3}$ is irrational, given that $\sqrt{3}$ is irrational. 3

**17.** Find the HCF and LCM of 336 and 54 by prime factorisation and verify HCF $\times$ LCM $=$ product. 3

OR

Prove that $\sqrt{7}$ is irrational. 3

**18.** Show that $7 + 1\sqrt{2}$ is irrational, given that $\sqrt{2}$ is irrational. 3

OR

Find the HCF and LCM of 144 and 198 by prime factorisation and verify HCF $\times$ LCM $=$ product. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** Show that the square of any positive integer is of the form $5q$, $5q + 1$ or $5q + 4$. 5

OR

Show that the cube of any positive integer is of the form $9m$, $9m + 1$ or $9m + 8$. 5

**20.** Use Euclid's division algorithm to find the HCF of 455 and 42. 5

OR

Prove that $\sqrt{3}$ is irrational and hence that $5 - 2\sqrt{3}$ is irrational. 5

## END OF QUESTION PAPER — Total: 40 Marks

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